

Project Ethereum USD Historical Dataset

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Software Used:

- **pandas** - For data loading, manipulation, and CSV operations
- **numpy** - For numerical calculations and statistical computations
- **matplotlib.pyplot** - For creating all visualizations (histograms, box plots, scatter plots, etc.)
- **seaborn** - For enhanced statistical visualizations and correlation heatmap
- **scipy.stats** - For hypothesis testing (t-tests, correlation, chi-square) and probability plots
- **os** - For directory creation and file path management

Problem Statement:

The project performs a comprehensive statistical analysis of Ethereum USD historical price data. It calculates descriptive statistics (mean, mode, median, standard deviation, variance, quartiles, IQR, range), creates data visualizations (histograms, box plots, density plots, scatter plots, correlation heatmap, Q-Q plots), and performs hypothesis testing (one-sample t-test, two-sample t-test, correlation test, paired t-test, chi-square test) on the Ethereum dataset containing 3,006 records with 39 variables (33 numeric analyzed) from 2017-2026.